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Credit Derivatives

An important development in derivatives markets since the late 1990s has been the growth of credit derivatives. In 2000, the total notional principal for outstanding credit derivatives contracts was about \$800 billion. By December 2009, this had become \$32 trillion. Credit derivatives are contracts where the payoff depends on the creditworthiness of one or more companies or countries. This chapter explains how credit derivatives work and how they are valued.

Credit derivatives allow companies to trade credit risks in much the same way that they trade market risks. Banks and other financial institutions used to be in the position where they could do little once they had assumed a credit risk except wait (and hope for the best). Now they can actively manage their portfolios of credit risks, keeping some and entering into credit derivatives contracts to protect themselves from others. As indicated in Business Snapshot 24.1, banks have been the biggest buyers of credit protection and insurance companies have been the biggest sellers.

Credit derivatives can be categorized as "single-name" or "multi-name." The most popular single-name credit derivative is a credit default swap. The payoff from this instrument depends on the creditworthiness of one company or country. There are two sides to the contract: the buyer and seller of protection. There is a payoff from the seller of protection to the buyer of protection if the specified entity (company or country) defaults on its obligations. The most popular multi-name credit derivative is a collateralized debt obligation. In this, a portfolio of debt instruments is specified and a complex structure is created where the cash flows from the portfolio are channelled to different categories of investors. Chapter 8 describes how multi-name credit derivatives were created from residential mortgages during the period leading up to the credit crisis. This chapter focuses on the situation where the underlying credit risks are those of corporations or countries. Multi-name credit derivatives increased in popularity relative to single-name credit derivatives up to June 2007 but became less popular during the 2007–2009 credit crisis.

This chapter starts by explaining how credit default swaps work and how they are valued. It then covers the trading of forwards and options on credit default swaps and total return swaps. It explains credit indices, basket credit default swaps, asset-backed securities, and collateralized debt obligations. It expands on the material in Chapter 23 to show how the Gaussian copula model of default correlation can be used to value tranches of collateralized debt obligations.

Business Snapshot 24.1 Who Bears the Credit Risk?

Traditionally banks have been in the business of making loans and then bearing the credit risk that the borrower will default. However, banks have for some time been reluctant to keep loans on their balance sheets. This is because, after the capital required by regulators has been accounted for, the average return earned on loans is often less attractive than that on other assets. As discussed in Section 8.1, banks created asset-backed securities to pass loans (and their credit risk) on to investors. In the late 1990s and early 2000s, banks also made extensive use of credit derivatives to shift the credit risk in their loans to other parts of the financial system.

If banks have been net buyers of credit protection, who have been net sellers? The answer is insurance companies. Insurance companies are not regulated in the same way as banks and as a result are sometimes more willing to bear credit risks than banks.

The result of all this is that the financial institution bearing the credit risk of a loan is often different from the financial institution that did the original credit checks. As the credit crisis of 2007 has shown, this is not always good for the overall health of the financial system.

24.1 CREDIT DEFAULT SWAPS

The most popular credit derivative is a *credit default swap* (CDS). This is a contract that provides insurance against the risk of a default by particular company. The company is known as the *reference entity* and a default by the company is known as a *credit event*. The buyer of the insurance obtains the right to sell bonds issued by the company for their face value when a credit event occurs and the seller of the insurance agrees to buy the bonds for their face value when a credit event occurs.[1] The total face value of the bonds that can be sold is known as the credit default swap's *notional principal*.

The buyer of the CDS makes periodic payments to the seller until the end of the life of the CDS or until a credit event occurs. These payments are typically made in arrears every quarter, but deals where payments are made every month, 6 months, or 12 months also occur and sometimes payments are made in advance. The settlement in the event of a default involves either physical delivery of the bonds or a cash payment.

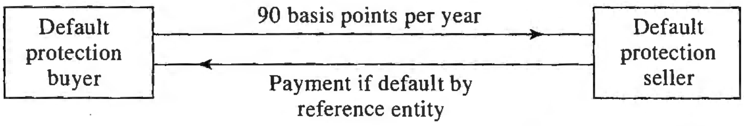
An example will help to illustrate how a typical deal is structured. Suppose that two parties enter into a 5-year credit default swap on March 20, 2012. Assume that the notional principal is \$100 million and the buyer agrees to pay 90 basis points per annum for protection against default by the reference entity, with payments being made quarterly in arrears.

The CDS is shown in Figure 24.1. If the reference entity does not default (i.e., there is no credit event), the buyer receives no payoff and pays 22.5 basis points (a quarter of 90 basis points) on \$100 million on June 20, 2012, and every quarter thereafter until March 20, 2017. The amount paid each quarter is $0.00225 \times 100{,}000{,}000$, or \$225,000.[2]

[1] The face value (or par value) of a coupon-bearing bond is the principal amount that the issuer repays at maturity if it does not default.

[2] The quarterly payments are liable to be slightly different from \$225,000 because of the application of the day count conventions described in Chapter 6.

Figure 24.1 Credit default swap.



If there is a credit event, a substantial payoff is likely. Suppose that the buyer notifies the seller of a credit event on May 20, 2015 (2 months into the fourth year). If the contract specifies physical settlement, the buyer has the right to sell bonds issued by the reference entity with a face value of $100 million for $100 million. If, as is now usual, there is cash settlement, an ISDA-organized auction process is used to determine the mid-market value of the cheapest deliverable bond several days after the credit event. Suppose the auction indicates that the bond is worth $35 per $100 of face value. The cash payoff would be $65 million.

The regular quarterly, semiannual, or annual payments from the buyer of protection to the seller of protection cease when there is a credit event. However, because these payments are made in arrears, a final accrual payment by the buyer is usually required. In our example, where there is a default on May 20, 2015, the buyer would be required to pay to the seller the amount of the annual payment accrued between March 20, 2015, and May 20, 2015 (approximately $150,000), but no further payments would be required.

The total amount paid per year, as a percent of the notional principal, to buy protection (90 basis points in our example) is known as the *CDS spread.* Several large banks are market makers in the credit default swap market. When quoting on a new 5-year credit default swap on a company, a market maker might bid 250 basis points and offer 260 basis points. This means that the market maker is prepared to buy protection by paying 250 basis points per year (i.e., 2.5% of the principal per year) and to sell protection for 260 basis points per year (i.e., 2.6% of the principal per year).

Many different companies and countries are reference entities for the CDS contracts that trade. As mentioned, payments are usually made quarterly in arrears. Contracts with maturities of 5 years are most popular, but other maturities such as 1, 2, 3, 7, and 10 years are not uncommon. Usually contracts mature on one of the following standard dates: March 20, June 20, September 20, and December 20. The effect of this is that the actual time to maturity of a contract when it is initiated is close to, but not necessarily the same as, the number of years to maturity that is specified. Suppose you call a dealer on November 15, 2012, to buy 5-year protection on a company. The contract would probably last until December 20, 2017. Your first payment would be due on December 20, 2012, and would equal an amount covering the November 15, 2012, to December 20, 2012, period.[3] A key aspect of a CDS contract is the definition of a credit event (i.e., a default). Usually a credit event is defined as a failure to make a payment as it becomes due, a restructuring of debt, or a bankruptcy. Restructuring is sometimes excluded in North American contracts, particularly in situations where the yield on the company's debt is high. More information on the CDS market is given in Business Snapshot 24.2.

[3] If the time to the first standard date is less than 1 month, then the first payment is typically made on the second standard payment date; otherwise it is made on the first standard payment date.

Business Snapshot 24.2 The CDS Market

In 1998 and 1999, the International Swaps and Derivatives Association (ISDA) developed a standard contract for trading credit default swaps in the over-the-counter market. Since then the market has grown very fast. A CDS contract is like an insurance contract in many ways, but there is one key difference. An insurance contract provides protection against losses on an asset that is owned. In the case of a CDS, the underlying asset does not have to be owned.

During the credit turmoil that started in August 2007, regulators became very concerned about systemic risk (see Business Snapshot 2.3). They felt that credit default swaps were a source of vulnerability for financial markets. The danger is that a default by one financial institution might lead to big losses by its counterparties in CDS transactions and further defaults by other financial institutions. Regulatory concerns were fueled by troubles at insurance giant AIG. This was a big seller of protection on the AAA-rated tranches created from mortgages (see Chapter 8). The protection proved very costly to AIG and the company was bailed out by the U.S. government.

Regulatory concerns have led to the development of clearing houses for CDS trades (and other over-the-counter derivatives) between financial institutions. The clearing house requires market participants to post margin on their CDS trades in much the same way that traders post margin on futures contracts.

During 2007 and 2008, trading ceased in many types of credit derivatives, but CDSs continued to trade actively (although the cost of protection increased dramatically). The advantage of CDSs over some other credit derivatives is that the way they work is straightforward. Other credit derivatives, such as those created from the securitization of household mortgages (see Chapter 8), lack this transparency.

It is not uncommon for the volume of CDSs on a company to be greater than its debt. Cash settlement of contracts is then clearly necessary. When Lehman defaulted in September 2008, there was about $400 billion of CDS contracts and $155 billion of Lehman debt outstanding. The payout to the buyers of protection (determined by an auction process) was 91.375% of principal.

Credit Default Swaps and Bond Yields

A CDS can be used to hedge a position in a corporate bond. Suppose that an investor buys a 5-year corporate bond yielding 7% per year for its face value and at the same time enters into a 5-year CDS to buy protection against the issuer of the bond defaulting. Suppose that the CDS spread is 200 basis points, or 2%, per annum. The effect of the CDS is to convert the corporate bond to a risk-free bond (at least approximately). If the bond issuer does not default the investor earns 5% per year when the CDS spread is netted against the corporate bond yield. If the bond does default the investor earns 5% up to the time of the default. Under the terms of the CDS, the investor is then able to exchange the bond for its face value. This face value can be invested at the risk-free rate for the remainder of the 5 years.

This shows that the excess of an n-year bond yield over the risk-free rate should approximately equal the n-year CDS spread. If it is markedly more than this, an investor can earn more than the risk-free rate by buying the corporate bond and buying protection. If it is markedly less than this, an investor can borrow at less than the risk-free rate by shorting the corporate bond and selling CDS protection. The relevant risk-free rate is usually assumed to be the LIBOR/swap rate, so that the excess of the bond yield over the risk-free rate is the asset swap spread (see Section 23.4).

The CDS–bond basis is defined as

CDS–bond basis = CDS spread – Excess of bond yield over risk-free rate

or equivalently

CDS–bond basis = CDS spread – Asset swap spread

The arbitrage argument given above suggests that this should be close to zero. Prior to the 2007 credit crisis, it was on average slightly positive. During the crisis, it tended to be negative and became highly negative for a short period of time in January 2009.

The Cheapest-to-Deliver Bond

As explained in Section 23.3, the recovery rate on a bond is defined as the value of the bond immediately after default as a percent of face value. This means that the payoff from a CDS is $L(1 - R)$, where L is the notional principal and R is the recovery rate.

Usually a CDS specifies that a number of different bonds can be delivered in the event of a default. The bonds typically have the same seniority, but they may not sell for the same percentage of face value immediately after a default.[4] This gives the holder of a CDS a cheapest-to-deliver bond option. As already mentioned, an auction process, organized by ISDA, is usually used to determine the value of the cheapest-to-deliver bond and, therefore, the payoff to the buyer of protection.

24.2 VALUATION OF CREDIT DEFAULT SWAPS

The CDS spread for a particular reference entity can be calculated from default probability estimates. We will illustrate how this is done with a simple example.

Suppose that the probability of a reference entity defaulting during a year conditional on no earlier default is 2%.[5] Table 24.1 shows survival probabilities and unconditional default probabilities (i.e., default probabilities as seen at time zero) for each of the 5 years. The probability of a default during the first year is 0.02 and the probability the reference entity will survive until the end of the first year is 0.98. The probability of a default during the second year is $0.02 \times 0.98 = 0.0196$ and the probability of survival until the end of the second year is $0.98 \times 0.98 = 0.9604$. The probability of default during the third year is $0.02 \times 0.9604 = 0.0192$, and so on.

We will assume that defaults always happen halfway through a year and that payments on the credit default swap are made once a year, at the end of each year. We also assume that the risk-free (LIBOR) interest rate is 5% per annum with continuous compounding

[4] There are a number of reasons for this. The claim that is made in the event of a default is typically equal to the bond's face value plus accrued interest. Bonds with high accrued interest at the time of default therefore tend to have higher prices immediately after default. Also the market may judge that in the event of a reorganization of the company some bond holders will fare better than others.

[5] This is a hazard rate expressed with annual compounding. The equivalent continuously compounded hazard rate is 2.02%.

Table 24.1 Unconditional default probabilities and survival probabilities.

Time (years)	*Default probability*	*Survival probability*
1	0.0200	0.9800
2	0.0196	0.9604
3	0.0192	0.9412
4	0.0188	0.9224
5	0.0184	0.9039

and the recovery rate is 40%. There are three parts to the calculation. These are shown in Tables 24.2, 24.3, and 24.4.

Table 24.2 shows the calculation of the present value of the expected payments made on the CDS assuming that payments are made at the rate of s per year and the notional principal is \$1. For example, there is a 0.9412 probability that the third payment of s is made. The expected payment is therefore $0.9412s$ and its present value is $0.9412se^{-0.05\times 3} = 0.8101s$. The total present value of the expected payments is $4.0704s$.

Table 24.3 shows the calculation of the present value of the expected payoff assuming a notional principal of \$1. As mentioned earlier, we are assuming that defaults always happen halfway through a year. For example, there is a 0.0192 probability of a payoff halfway through the third year. Given that the recovery rate is 40%, the expected payoff at this time is $0.0192 \times 0.6 \times 1 = 0.0115$. The present value of the expected payoff is $0.0115e^{-0.05\times 2.5} = 0.0102$. The total present value of the expected payoffs is \$0.0511.

As a final step, Table 24.4 considers the accrual payment made in the event of a default. For example, there is a 0.0192 probability that there will be a final accrual payment halfway through the third year. The accrual payment is $0.5s$. The expected accrual payment at this time is therefore $0.0192 \times 0.5s = 0.0096s$. Its present value is $0.0096se^{-0.05\times 2.5} = 0.0085s$. The total present value of the expected accrual payments is $0.0426s$.

Table 24.2 Calculation of the present value of expected payments. Payment $= s$ per annum.

Time (years)	*Probability of survival*	*Expected payment*	*Discount factor*	*PV of expected payment*
1	0.9800	$0.9800s$	0.9512	$0.9322s$
2	0.9604	$0.9604s$	0.9048	$0.8690s$
3	0.9412	$0.9412s$	0.8607	$0.8101s$
4	0.9224	$0.9224s$	0.8187	$0.7552s$
5	0.9039	$0.9039s$	0.7788	$0.7040s$
Total				$4.0704s$

Table 24.3 Calculation of the present value of expected payoff. Notional principal = \$1.

Time (years)	*Probability of default*	*Recovery rate*	*Expected payoff (\$)*	*Discount factor*	*PV of expected payoff (\$)*
0.5	0.0200	0.4	0.0120	0.9753	0.0117
1.5	0.0196	0.4	0.0118	0.9277	0.0109
2.5	0.0192	0.4	0.0115	0.8825	0.0102
3.5	0.0188	0.4	0.0113	0.8395	0.0095
4.5	0.0184	0.4	0.0111	0.7985	0.0088
Total					0.0511

From Tables 24.2 and 24.4, the present value of the expected payments is

$$4.0704s + 0.0426s = 4.1130s$$

From Table 24.3, the present value of the expected payoff is 0.0511. Equating the two gives

$$4.1130s = 0.0511$$

or $s = 0.0124$. The mid-market CDS spread for the 5-year deal we have considered should be 0.0124 times the principal or 124 basis points per year. This result can also be produced using the DerivaGem CDS worksheet. The hazard rate (continuously compounded in DerivaGem) should be input as 2.02% for all maturities, the term structure is flat at 5%, and the recovery rate is 40%.

The calculations assume that defaults happen only at points midway between payment dates. This simple assumption can be relaxed, but usually gives good results.

Marking to Market a CDS

A CDS, like most other swaps, is marked to market daily. It may have a positive or negative value. Suppose, for example the credit default swap in our example had been negotiated some time ago for a spread of 150 basis points, the present value of the payments by the buyer would be $4.1130 \times 0.0150 = 0.0617$ and the present value of the

Table 24.4 Calculation of the present value of accrual payment.

Time (years)	*Probability of default*	*Expected accrual payment*	*Discount factor*	*PV of expected accrual payment*
0.5	0.0200	0.0100s	0.9753	0.0097s
1.5	0.0196	0.0098s	0.9277	0.0091s
2.5	0.0192	0.0096s	0.8825	0.0085s
3.5	0.0188	0.0094s	0.8395	0.0079s
4.5	0.0184	0.0092s	0.7985	0.0074s
Total				0.0426s

payoff would be 0.0511 as above. The value of swap to the seller would therefore be $0.0617 - 0.0511$, or 0.0106 times the principal. Similarly the mark-to-market value of the swap to the buyer of protection would be -0.0106 times the principal.

Estimating Default Probabilities

The default probabilities used to value a CDS should be risk-neutral default probabilities, not real-world default probabilities (see Section 23.5 for a discussion of the difference between the two). Risk-neutral default probabilities can be estimated from bond prices or asset swaps as explained in Chapter 23. An alternative is to imply them from CDS quotes. The latter approach is similar to the practice in options markets of implying volatilities from the prices of actively traded options.

Suppose we change the example in Tables 24.2, 24.3 and 24.4 so that we do not know the default probabilities. Instead we know that the mid-market CDS spread for a newly issued 5-year CDS is 100 basis points per year. We can reverse-engineer our calculations (using Excel in conjunction with Solver) to conclude that the implied default probability per year (conditional on no earlier default) is 1.61% per year.[6]

Binary Credit Default Swaps

A binary credit default swap is structured similarly to a regular credit default swap except that the payoff is a fixed dollar amount. Suppose that, in the example we considered in Tables 24.1 to 24.4, the payoff is \$1 instead of $1 - R$ dollars and the swap spread is s. Tables 24.1, 24.2 and 24.4 are the same, but Table 24.3 is replaced by Table 24.5. The CDS spread for a new binary CDS is given by $4.1130s = 0.0852$, so that the CDS spread, s, is 0.0207, or 207 basis points.

How Important is the Recovery Rate?

Whether we use CDS spreads or bond prices to estimate default probabilities we need an estimate of the recovery rate. However, provided that we use the same recovery rate

Table 24.5 Calculation of the present value of expected payoff from a binary credit default swap. Principal = \$1.

Time (years)	*Probability of default*	*Expected payoff (\$)*	*Discount factor*	*PV of expected payoff (\$)*
0.5	0.0200	0.0200	0.9753	0.0195
1.5	0.0196	0.0196	0.9277	0.0182
2.5	0.0192	0.0192	0.8825	0.0170
3.5	0.0188	0.0188	0.8395	0.0158
4.5	0.0184	0.0184	0.7985	0.0147
Total				0.0852

[6] The DerivaGem worksheet gives a continuously compounded hazard rate of 1.626%. This is equivalent to 1.61% with annual compounding. If spreads for CDS swaps with different maturities are available, DerivaGem calculates a step function for the hazard rate.

for (a) estimating risk-neutral default probabilities and (b) valuing a CDS, the value of the CDS (or the estimate of the CDS spread) is not very sensitive to the recovery rate. This is because the implied probabilities of default are approximately proportional to $1/(1-R)$ and the payoffs from a CDS are proportional to $1-R$.

This argument does not apply to the valuation of binary CDS. Implied probabilities of default are still approximately proportional to $1/(1-R)$. However, for a binary CDS, the payoffs from the CDS are independent of R. If we have a CDS spread for both a plain vanilla CDS and a binary CDS, we can estimate both the recovery rate and the default probability (see Problem 24.25).

The Future of the CDS Market

The credit default swap market survived the credit crunch of 2007 reasonably well. It is true that it has come under a great deal of regulatory scrutiny and CDSs are being moved to clearing houses. But their importance is unlikely to decline. They are important tools for managing credit risk. A financial institution can reduce its credit exposure to particular companies by buying protection. It can also use CDSs to diversify credit risk. For example, if a financial institution has too much credit exposure to a particular business sector, it can buy protection against defaults by companies in the sector and at the same time sell protection against default by companies in other unrelated sectors.

Some market participants think the CDS market will eventually be as big as the interest rate swap market. Others are less optimistic. There is a potential asymmetric information problem in the CDS market that is not present in other over-the-counter derivatives markets (see Business Snapshot 24.3).

24.3 CREDIT INDICES

Participants in credit markets have developed indices to track credit default swap spreads. In 2004 there were agreements between different producers of indices that led to some consolidation. Two important standard portfolios used by index providers are:

1. CDX NA IG, a portfolio of 125 investment grade companies in North America
2. iTraxx Europe, a portfolio of 125 investment grade names in Europe

These portfolios are updated on March 20 and September 20 each year. Companies that are no longer investment grade are dropped from the portfolios and new investment grade companies are added.[7]

Suppose that the 5-year CDX NA IG index is quoted by a market maker as bid 65 basis points, offer 66 basis points. (This is referred to as the index spread.) Roughly speaking, this means that a trader can buy CDS protection on all 125 companies in the index for 66 basis points per company. Suppose a trader wants \$800,000 of protection on each company. The total cost is $0.0066 \times 800{,}000 \times 125$, or \$660,000 per year. The trader can similarly sell \$800,000 of protection on each of the 125 companies for a total of \$650,000 per annum. When a company defaults, the protection buyer receives the

[7] On September 20, 2010, the Series 14 iTraxx Europe portfolio and the Series 15 CDX NA IG portfolio were defined. The series numbers indicate that, by the end of September 2010, the iTraxx Europe portfolio had been updated 13 times and the CDX NA IG portfolio had been updated 14 times.

Business Snapshot 24.3 Is the CDS Market a Fair Game?

There is one important difference between credit default swaps and the other over-the-counter derivatives that we have considered in this book. The other over-the-counter derivatives depend on interest rates, exchange rates, equity indices, commodity prices, and so on. There is no reason to assume that any one market participant has better information than any other market participant about these variables.

Credit default swaps spreads depend on the probability that a particular company will default during a particular period of time. Arguably some market participants have more information to estimate this probability than others. A financial institution that works closely with a particular company by providing advice, making loans, and handling new issues of securities is likely to have more information about the creditworthiness of the company than another financial institution that has no dealings with the company. Economists refer to this as an *asymmetric information* problem.

Whether asymmetric information will curtail the expansion of the credit default swap market remains to be seen. Financial institutions emphasize that the decision to buy protection against the risk of default by a company is normally made by a risk manager and is not based on any special information that may exist elsewhere in the financial institution about the company.

usual CDS payoff and the annual payment is reduced by 660,000/125 = \$5,280. There is an active market in buying and selling CDS index protection for maturities of 3, 5, 7, and 10 years. The maturities for these types of contracts on the index are usually December 20 and June 20. (This means that a "5-year" contract actually lasts between $4\frac{3}{4}$ and $5\frac{1}{4}$ years.) Roughly speaking, the index is the average of the CDS spreads on the companies in the underlying portfolio.[8]

24.4 THE USE OF FIXED COUPONS

The precise way in which CDS and CDS index transactions work is a little more complicated than has been described up to now. For each underlying and each maturity, a coupon and a recovery rate are specified. A price is calculated from the quoted spread using the following procedure:

1. Assume four payments per year, made in arrears.
2. Imply a hazard rate (default intensity) from the quoted spread. This involves calculations similar to those in Section 24.2. An iterative search is used to determine the hazard rate that leads to the quoted spread.

[8] More precisely, the index is slightly lower than the average of the credit default swap spreads for the companies in the portfolio. To understand the reason for this consider a portfolio consisting of two companies, one with a spread of 1,000 basis points and the other with a spread of 10 basis points. To buy protection on the companies would cost slightly less than 505 basis points per company. This is because the 1,000 basis points is not expected to be paid for as long as the 10 basis points and should therefore carry less weight. Another complication for CDX NA IG, but not iTraxx Europe, is that the definition of default applicable to the index includes restructuring, whereas the definition for CDS contracts on the underlying companies often does not.

3. Calculate a duration D for the CDS payments. This is the number that the spread is multiplied by to get the present value of the spread payments. (In the example in Section 24.2, it is 4.1130.)
4. The price P is given by $P = 100 - 100 \times D \times (S - C)$, where S is the spread and C is the coupon expressed in decimal form.

When a trader buys protection the trader pays $100 - P$ per \$100 of the total remaining notional and the seller of protection receives this amount. (If $100 - P$ is negative, the buyer of protection receives money and the seller of protection pays money.) The buyer of protection then pays the coupon times the remaining notional on each payment date. (On a CDS, the remaining notional is the original notional until default and zero thereafter. For a CDS index, the remaining notional is the number of names in the index that have not yet defaulted multiplied by the principal per name.) The payoff when there is a default is calculated in the usual way. This arrangement facilitates trading because the regular quarterly payments made by the buyer of protection are independent of the spread at the time the buyer enters into the contract.

Example 24.1

Suppose that the iTraxx Europe index quote is 34 basis points and the coupon is 40 basis points for a contract lasting exactly 5 years, with both quotes being expressed using a 30/360 day count. (This is the usual day count convention in CDS and CDS index markets.) The equivalent actual/actual quotes are 0.345% for the index and 0.406% for the coupon. Suppose that the yield curve is flat at 4% per year (actual/actual, continuously compounded). The specified recovery rate is 40%. With four payments per year in arrears, the implied hazard rate is 0.5717%. The duration is 4.447 years. The price is therefore

$$100 - 100 \times 4.447 \times (0.00345 - 0.00406) = 100.27$$

Consider a contract where protection is \$1 million per name. Initially, the seller of protection would pay the buyer $\$1{,}000{,}000 \times 125 \times 0.0027$. Thereafter, the buyer of protection would make quarterly payments in arrears at an annual rate of $\$1{,}000{,}000 \times 0.00406 \times n$, where n is the number of companies that have not defaulted. When a company defaults, the payoff is calculated in the usual way and there is an accrual payment from the buyer to the seller calculated at the rate of 0.406% per year on \$1 million.

24.5 CDS FORWARDS AND OPTIONS

Once the CDS market was well established, it was natural for derivatives dealers to trade forwards and options on credit default swap spreads.[9]

A forward credit default swap is the obligation to buy or sell a particular credit default swap on a particular reference entity at a particular future time T. If the reference entity defaults before time T, the forward contract ceases to exist. Thus a bank could enter into a forward contract to sell 5-year protection on a company for

[9] The valuation of these instruments is discussed in J.C. Hull and A. White, "The Valuation of Credit Default Swap Options," *Journal of Derivatives*, 10, 5 (Spring 2003): 40–50.

280 basis points starting in 1 year. If the company defaulted before the 1-year point, the forward contract would cease to exist.

A credit default swap option is an option to buy or sell a particular credit default swap on a particular reference entity at a particular future time T. For example, a trader could negotiate the right to buy 5-year protection on a company starting in 1 year for 280 basis points. This is a call option. If the 5-year CDS spread for the company in 1 year turns out to be more than 280 basis points, the option will be exercised; otherwise it will not be exercised. The cost of the option would be paid up front. Similarly an investor might negotiate the right to sell 5-year protection on a company for 280 basis points starting in 1 year. This is a put option. If the 5-year CDS spread for the company in 1 year turns out to be less than 280 basis points, the option will be exercised; otherwise it will not be exercised. Again the cost of the option would be paid up front. Like CDS forwards, CDS options are usually structured so that they cease to exist if the reference entity defaults before option maturity.

24.6 BASKET CREDIT DEFAULT SWAPS

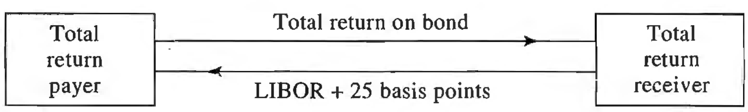
In what is referred to as a *basket credit default swap* there are a number of reference entities. An *add-up basket* CDS provides a payoff when any of the reference entities default. A *first-to-default* CDS provides a payoff only when the first default occurs. A *second-to-default* CDS provides a payoff only when the second default occurs. More generally, a *kth-to-default* CDS provides a payoff only when the *k*th default occurs. Payoffs are calculated in the same way as for a regular CDS. After the relevant default has occurred, there is a settlement. The swap then terminates and there are no further payments by either party.

24.7 TOTAL RETURN SWAPS

A *total return swap* is a type of credit derivative. It is an agreement to exchange the total return on a bond (or any portfolio of assets) for LIBOR plus a spread. The total return includes coupons, interest, and the gain or loss on the asset over the life of the swap.

An example of a total return swap is a 5-year agreement with a notional principal of $100 million to exchange the total return on a corporate bond for LIBOR plus 25 basis points. This is illustrated in Figure 24.2. On coupon payment dates the payer pays the coupons earned on an investment of $100 million in the bond. The receiver pays interest at a rate of LIBOR plus 25 basis points on a principal of $100 million. (LIBOR is set on one coupon date and paid on the next as in a plain vanilla interest rate swap.) At the

Figure 24.2 Total return swap.



end of the life of the swap there is a payment reflecting the change in value of the bond. For example, if the bond increases in value by 10% over the life of the swap, the payer is required to pay \$10 million (= 10% of \$100 million) at the end of the 5 years. Similarly, if the bond decreases in value by 15%, the receiver is required to pay \$15 million at the end of the 5 years. If there is a default on the bond, the swap is usually terminated and the receiver makes a final payment equal to the excess of \$100 million over the market value of the bond.

If the notional principal is added to both sides at the end of the life of the swap, the total return swap can be characterized as follows. The payer pays the cash flows on an investment of \$100 million in the corporate bond. The receiver pays the cash flows on a \$100 million bond paying LIBOR plus 25 basis points. If the payer owns the corporate bond, the total return swap allows it to pass the credit risk on the bond to the receiver. If it does not own the bond, the total return swap allows it to take a short position in the bond.

Total return swaps are often used as a financing tool. One scenario that could lead to the swap in Figure 24.2 is as follows. The receiver wants financing to invest \$100 million in the reference bond. It approaches the payer (which is likely to be a financial institution) and agrees to the swap. The payer then invests \$100 million in the bond. This leaves the receiver in the same position as it would have been if it had borrowed money at LIBOR plus 25 basis points to buy the bond. The payer retains ownership of the bond for the life of the swap and faces less credit risk than it would have done if it had lent money to the receiver to finance the purchase of the bond, with the bond being used as collateral for the loan. If the receiver defaults the payer does not have the legal problem of trying to realize on the collateral. Total return swaps are similar to repos (see Section 4.1) in that they are structured to minimize credit risk when securities are being financed.

The spread over LIBOR received by the payer is compensation for bearing the risk that the receiver will default. The payer will lose money if the receiver defaults at a time when the reference bond's price has declined. The spread therefore depends on the credit quality of the receiver, the credit quality of the bond issuer, and the correlation between the two.

There are a number of variations on the standard deal we have described. Sometimes, instead of there being a cash payment for the change in value of the bond, there is physical settlement where the payer exchanges the underlying asset for the notional principal at the end of the life of the swap. Sometimes the change-in-value payments are made periodically rather than all at the end.

24.8 COLLATERAL DEBT OBLIGATIONS

We discussed asset-backed securities (ABSs) in Chapter 8. Figure 8.1 shows a simple structure. An ABS where the underlying assets are bonds is known as a *collateralized debt obligation*, or *CDO*. A waterfall similar to that indicated in Figure 8.2 is defined for the interest and principal payments on the bonds. The precise rules underlying the waterfall are complicated, but they are designed to ensure that if one tranche is more senior than another it is more likely to receive promised interest payments and repayments of principal.

Synthetic CDOs

When a CDO is created from a bond portfolio, as just described, the resulting structure is known as a *cash CDO*. In an important market development, it was recognized that a long position in a corporate bond has a similar risk to a short position in a CDS when the reference entity in the CDS is the company issuing the bond. This led an alternative structure known as a *synthetic CDO*, which has become very popular.

The originator of a synthetic CDO chooses a portfolio of companies and a maturity (e.g., 5 years) for the structure. It sells CDS protection on each company in the portfolio with the CDS maturities equaling the maturity of the structure. The synthetic CDO principal is the total of the notional principals underlying the CDSs. The originator has cash inflows equal to the the CDS spreads and cash outflows when companies in the portfolio default. Tranches are formed and the cash inflows and outflows are distributed to tranches. The rules for determining the cash inflows and outflows of tranches are more straightforward for a synthetic CDO than for a cash CDO. Suppose that there are only three tranches: equity, mezzanine, and senior. The rules might be as follows:

1. The equity tranche is responsible for the payouts on the CDSs until they reach 5% of the synthetic CDO principal. It earns a spread of 1000 basis points per year on the outstanding tranche principal.
2. The mezzanine tranche is responsible for payouts in excess of 5% up to a maximum of 20% of the synthetic CDO principal. It earns a spread of 100 basis points per year on the outstanding tranche principal.
3. The senior tranche is responsible for payouts in excess of 20%. It earns a spread of 10 basis points per year on the outstanding tranche principal.

To understand how the synthetic CDO would work, suppose that its principal is \$100 million. The equity, mezzanine, and senior tranche principals are \$5 million, \$15 million, and \$80 million, respectively. The tranches initially earn the specified spreads on these notional principals. Suppose that after 1 year defaults by companies in the portfolio lead to payouts of \$2 million on the CDSs. The equity tranche holders are responsible for these payouts. The equity tranche principal reduces to \$3 million and its spread (1,000 basis points) is then earned on \$3 million instead of \$5 million. If, later during the life of the CDO, there are further payouts of \$4 million on the CDSs, the cumulative of the payments required by the equity tranche is \$5 million, so that its outstanding principal becomes zero. The mezzanine tranche holders have to pay \$1 million. This reduces their outstanding principal to \$14 million.

Cash CDOs require an initial investment by the tranche holders (to finance the underlying bonds). By contrast, the holders of synthetic CDOs do not have to make an initial investment. They just have to agree to the way cash inflows and outflows will be calculated. In practice, they are almost invariably required to post the initial tranche principal as collateral. When the tranche becomes responsible for a payoff on a CDS, the money is taken out of the collateral. The balance in the collateral account earns interest at LIBOR.

Standard Portfolios and Single-Tranche Trading

In the synthetic CDO we have described, the tranche holders sell protection to the originator of the CDO, who in turn sells protection on CDSs to other market

Table 24.6 Mid-market quotes for 5-year tranches of iTraxx Europe. Quotes are in basis points except for the 0–3% tranche where the quote equals the percent of the tranche principal that must be paid up front in addition to 500 basis points per year. (*Source*: Creditex Group Inc.)

	Tranche					*iTraxx*
Date	0–3%	3–6%	6–9%	9–12%	12–22%	*index*
January 31, 2007	10.34%	41.59	11.95	5.60	2.00	23
January 31, 2008	30.98%	316.90	212.40	140.00	73.60	77
January 30, 2009	64.28%	1185.63	606.69	315.63	97.13	165

participants. An innovation in the market was the trading of a tranche without the underlying portfolio of short CDS positions being created. This is sometimes referred to as *single-tranche trading*. There are two parties to a trade: the buyer of protection on a tranche and the seller of protection on the tranche. The portfolio of short CDS positions is used as a reference point to define the cash flows between the two sides, but it is not created. The buyer of protection pays the tranche spread to the seller of protection, and the seller of protection pays amounts to the buyer that correspond to those losses on the reference portfolio of CDSs that the tranche is responsible for.

In Section 24.3, we discussed CDS indices such as CDX NA IG and iTraxx Europe. The market has used the portfolios underlying these indices to define standard synthetic CDO tranches. These trade very actively. The six standard tranches of CDX NA IG cover losses in the ranges 0–3%, 3–6%, 6–9%, 9–12%, 12–22%, and 22–100%. The six standard tranches of iTraxx Europe cover losses in the ranges 0–3%, 3–7%, 7–10%, 10–15%, 15–30%, and 30–100%.

Table 24.6 shows the quotes for 5-year iTraxx tranches at the end of January of three successive years. The index spread is the cost in basis points of buying protection on all the companies in the index, as described in Section 24.3. The quotes for all tranches except the 0–3% tranche is the cost in basis point per year of buying tranche protection. (As explained earlier, this is paid on a principal that declines as the tranche experiences losses.) In the case of the 0–3% (equity) tranche, the protection buyer makes an initial payment and then pays 500 basis points per year on the outstanding tranche principal. The quote is for the initial payment as a percentage of the initial tranche principal.

What a difference two years makes in the credit markets! Table 24.6 shows that the credit crunch led to a huge increase in credit spreads. The iTraxx index rose from 23 basis points in January 2007 to 165 basis points in January 2009. The individual tranche quotes have also shown huge increases. One reason for the changes is that the market's assessment of default probabilities for investment-grade corporations has increased. However, it is also the case that protection sellers were in many cases experiencing liquidity problems. They became more averse to risk and increased the risk premiums they required.

24.9 ROLE OF CORRELATION IN A BASKET CDS AND CDO

The cost of protection in a kth-to-default CDS or a tranche of a CDO is critically dependent on default correlation. Suppose that a basket of 100 reference entities is used to define a 5-year kth-to-default CDS and that each reference entity has a risk-neutral probability of 2% of defaulting during the 5 years. When the default correlation between the reference entities is zero the binomial distribution shows that the probability of one or more defaults during the 5 years is 86.74% and the probability of 10 or more defaults is 0.0034%. A first-to-default CDS is therefore quite valuable whereas a tenth-to-default CDS is worth almost nothing.

As the default correlation increases the probability of one or more defaults declines and the probability of 10 or more defaults increases. In the limit where the default correlation between the reference entities is perfect the probability of one or more defaults equals the probability of ten or more defaults and is 2%. This is because in this extreme situation the reference entities are essentially the same. Either they all default (with probability 2%) or none of them default (with probability 98%).

The valuation of a tranche of a synthetic CDO is similarly dependent on default correlation. If the correlation is low, the junior equity tranche is very risky and the senior tranches are very safe. As the default correlation increases, the junior tranches become less risky and the senior tranches become more risky. In the limit where the default correlation is perfect and the recovery rate is zero, the tranches are equally risky.

24.10 VALUATION OF A SYNTHETIC CDO

Synthetic CDOs can be valued using the DerivaGem software. To explain the calculations, suppose that the payment dates on a synthetic CDO tranche are at times $\tau_1, \tau_2, \dots, \tau_m$ and $\tau_0 = 0$. Define E_j as the expected tranche principal at time τ_j and $v(\tau)$ as the present value of \$1 received at time τ. Suppose that the spread on a particular tranche (i.e., the number of basis points paid for protection) is s per year. This spread is paid on the remaining tranche principal. The present value of the expected regular spread payments on the CDO is therefore given by sA, where

$$A = \sum_{j=1}^{m} (\tau_j - \tau_{j-1}) E_j v(\tau_j) \tag{24.1}$$

The expected loss between times τ_{j-1} and τ_j is $E_{j-1} - E_j$. Assume that the loss occurs at the midpoint of the time interval (i.e., at time $0.5\tau_{j-1} + 0.5\tau_j$). The present value of the expected payoffs on the CDO tranche is

$$C = \sum_{j=1}^{m} (E_{j-1} - E_j) v(0.5\tau_{j-1} + 0.5\tau_j) \tag{24.2}$$

The accrual payment due on the losses is given by sB, where

$$B = \sum_{j=1}^{m} 0.5(\tau_j - \tau_{j-1})(E_{j-1} - E_j) v(0.5\tau_{j-1} + 0.5\tau_j) \tag{24.3}$$

The value of the tranche to the protection buyer is $C - sA - sB$. The breakeven spread on the tranche occurs when the present value of the payments equals the present value of the payoffs or

$$C = sA + sB$$

The breakeven spread is therefore

$$s = \frac{C}{A + B} \tag{24.4}$$

Equations (24.1) to (24.3) show the key role played by the expected tranche principal in calculating the breakeven spread for a tranche. If we know the expected principal for a tranche on all payment dates and we also know the zero-coupon yield curve, the breakeven tranche spread can be calculated from equations (24.1) to (24.4).

Using the Gaussian Copula Model of Time to Default

The one-factor Gaussian copula model of time to default was introduced in Section 22.9. This is the standard market model for valuing synthetic CDOs. All companies are assumed to have the same probability $Q(t)$ of defaulting by time t. Equation (23.12) converts this unconditional probability of default by time t to the probability of default by time t conditional on the factor F:

$$Q(t \mid F) = N\left(\frac{N^{-1}[Q(t)] - \sqrt{\rho}\, F}{\sqrt{1-\rho}}\right) \tag{24.5}$$

Here ρ is the copula correlation, assumed to be the same for any pair of companies.

In the calculation of $Q(t)$, it is usually assumed that the hazard rate for a company is constant and consistent with the index spread. The hazard rate that is assumed can be calculated by using the CDS valuation approach in Section 24.2 and searching for the hazard rate that gives the index spread. Suppose that this hazard rate is λ. Then, from equation (23.1),

$$Q(t) = 1 - e^{-\lambda t} \tag{24.6}$$

From the properties of the binomial distribution, the standard market model gives the probability of exactly k defaults by time t, conditional on F, as

$$P(k, t \mid F) = \frac{n!}{(n-k)!\,k!} Q(t \mid F)^k [1 - Q(t \mid F)]^{n-k} \tag{24.7}$$

Suppose that the tranche under consideration covers losses on the portfolio between α_L and α_H. The parameter α_L is known as the *attachment point* and the parameter α_H is known as the *detachment point*. Define

$$n_L = \frac{\alpha_L n}{1-R} \quad \text{and} \quad n_H = \frac{\alpha_H n}{1-R}$$

where R is the recovery rate. Also, define $m(x)$ as the smallest integer greater than x. Without loss of generality, we assume that the initial tranche principal is 1. The tranche principal stays 1 while the number of defaults, k, is less than $m(n_L)$. It is zero when the number of defaults is greater than or equal to $m(n_H)$. Otherwise, the tranche principal is

$$\frac{\alpha_H - k(1-R)/n}{\alpha_H - \alpha_L}$$

Define $E_j(F)$ as the expected tranche principal at time τ_j conditional on the value of the factor F. It follows that

$$E_j(F) = \sum_{k=0}^{m(n_L)-1} P(k, \tau_j \mid F) + \sum_{k=m(n_L)}^{m(n_H)-1} P(k, \tau_j \mid F)\frac{\alpha_H - k(1-R)/n}{\alpha_H - \alpha_L} \tag{24.8}$$

Define $A(F)$, $B(F)$, and $C(F)$ as the values of A, B, and C conditional on F. Similarly to equations (24.1) to (24.3),

$$A(F) = \sum_{j=1}^{m}(\tau_j - \tau_{j-1})E_j(F)v(\tau_j) \tag{24.9}$$

$$B(F) = \sum_{j=1}^{m} 0.5(\tau_j - \tau_{j-1})(E_{j-1}(F) - E_j(F))v(0.5\tau_{j-1} + 0.5\tau_j) \tag{24.10}$$

$$C(F) = \sum_{j=1}^{m}(E_{j-1}(F) - E_j(F))v(0.5\tau_{j-1} + 0.5\tau_j) \tag{24.11}$$

The variable F has a standard normal distribution. To calculate the unconditional values of A, B, and C, it is necessary to integrate $A(F)$, $B(F)$, and $C(F)$ over a standard normal distribution. Once the unconditional values have been calculated, the breakeven spread on the tranche can be calculated as $C/(A+B)$.[10]

The integration is best accomplished with a procedure known as Gaussian quadrature. It involves the following approximation:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-F^2/2} g(F)\, dF \approx \sum_{k=1}^{k=M} w_k g(F_k) \tag{24.12}$$

As M increases, accuracy increases. The values of w_k and F_k for different values of M are given on the author's website.[11] The value of M is twice the "number of integration points" variable in DerivaGem. Setting the number of integration points equal to 20 usually gives good results.

Example 24.2

Consider the mezzanine tranche of iTraxx Europe (5-year maturity) when the copula correlation is 0.15 and the recovery rate is 40%. In this case, $\alpha_L = 0.03$, $\alpha_H = 0.06$, $n = 125$, $n_L = 6.25$, and $n_H = 12.5$. We suppose that the term structure of interest rates is flat at 3.5%, payments are made quarterly, and the CDS spread on the index is 50 basis points. A calculation similar to that in Section 24.2 shows that the constant hazard rate corresponding to the CDS spread is 0.83% (with continuous compounding). An extract from the remaining calculations is shown in Table 24.7. A value of $M = 60$ is used in equation (24.12). The factor values, F_k, and their weights, w_k, are shown in first segment of the table. The expected tranche principals on payment dates conditional on the factor values are calculated from equations (24.5) to (24.8)

[10] In the case of the equity tranche, the quote is the upfront payment that must be made in addition to 500 basis points per year. The breakeven upfront payment is $C - 0.05(A + B)$.

[11] The parameters w_k and F_k are calculated from the roots of Hermite polynomials. For more information on Gaussian quadrature, see Technical Note 21 at www.rotman.utoronto.ca/~hull/TechnicalNotes.

Table 24.7 Valuation of CDO in Example 24.2: principal = 1; payments are per unit of spread.

Weights and values for factors						
w_k	...	0.1579	0.1579	0.1342	0.0969	...
F_k	...	0.2020	−0.2020	−0.6060	−1.0104	...
Expected principal, $E_j(F_k)$						
Time						
$j = 1$	...	1.0000	1.0000	1.0000	1.0000	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮
$j = 19$	...	0.9953	0.9687	0.8636	0.6134	...
$j = 20$	...	0.9936	0.9600	0.8364	0.5648	...
PV expected payment, $A(F_k)$						
$j = 1$	...	0.2478	0.2478	0.2478	0.2478	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮
$j = 19$	...	0.2107	0.2051	0.1828	0.1299	...
$j = 20$	...	0.2085	0.2015	0.1755	0.1185	...
Total	...	4.5624	4.5345	4.4080	4.0361	...
PV expected accrual payment, $B(F_k)$						
$j = 1$	...	0.0000	0.0000	0.0000	0.0000	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮
$j = 19$	...	0.0001	0.0008	0.0026	0.0051	...
$j = 20$	...	0.0002	0.0009	0.0029	0.0051	...
Total	...	0.0007	0.0043	0.0178	0.0478	...
PV expected payoff, $C(F_k)$						
$j = 1$	...	0.0000	0.0000	0.0000	0.0000	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮
$j = 19$	...	0.0011	0.0062	0.0211	0.0412	...
$j = 20$	...	0.0014	0.0074	0.0230	0.0410	...
Total	...	0.0055	0.0346	0.1423	0.3823	...

and shown in the second segment of the table. The values of A, B, and C conditional on the factor values are calculated in the last three segments of the table using equations (24.9) to (24.11). The unconditional values of A, B, and C are calculated by integrating $A(F)$, $B(F)$, and $C(F)$ over the probability distribution of F. This is done by setting $g(F)$ equal in turn to $A(F)$, $B(F)$, and $C(F)$ in equation (24.12). The result is

$$A = 4.2846, \quad B = 0.0187, \quad C = 0.1496$$

The breakeven tranche spread is $0.1496/(4.2846 + 0.0187) = 0.0348$, or 348 basis points.

This result can be obtained from DerivaGem. The CDS worksheet is used to convert the 50-basis-point spread to a hazard rate of 0.83%. The CDO worksheet is then used with this hazard rate and 30 integration points.

Valuation of kth-to-Default CDS

A kth-to-default CDS (see Section 24.5) can also be valued using the standard market model by conditioning on the factor F. The conditional probability that the kth default happens between times τ_{j-1} and τ_j is the probability that there are k or more defaults by time τ_j minus the probability that there are k or more defaults by time τ_{j-1}. This can be calculated from equations (24.5) to (24.7) as

$$\sum_{q=k}^{n} P(q, \tau_j \mid F) - \sum_{q=k}^{n} P(q, \tau_{j-1} \mid F)$$

Defaults between time τ_{j-1} and τ_j can be assumed to happen at time $0.5\tau_{j-1} + 0.5\tau_j$. This allows the present value of payments and of payoffs, conditional on F, to be calculated in the same way as for regular CDS payoffs (see Section 24.2). By integrating over F, the unconditional present values of payments and payoffs can be calculated.

Example 24.3

Consider a portfolio consisting of 10 bonds each with the default probabilities in Table 24.1 and suppose we are interested in valuing a third-to-default CDS where payments are made annually in arrears. Assume that the copula correlation is 0.3, the recovery rate is 40%, and all risk-free rates are 5%. As in Table 24.7, we consider $M = 60$ different factor values. The unconditional cumulative probability of each bond defaulting by years 1, 2, 3, 4, 5 is 0.0200, 0.0396, 0.0588, 0.0776, 0.0961, respectively. Equation (24.5) shows that, conditional on $F = -1.0104$, these default probabilities are 0.0365, 0.0754, 0.1134, 0.1498, 0.1848, respectively. From the binomial distribution, the conditional probability of three or more defaults by times 1, 2, 3, 4, 5 years is 0.0048, 0.0344, 0.0950, 0.1794, 0.2767, respectively. The conditional probability of the third default happening during years 1, 2, 3, 4, 5 is therefore 0.0048, 0.296, 0.0606, 0.0844, 0.0974, respectively. An analysis similar to that in Section 24.2 shows that the present values of regular payments, accrual payments, and payoffs conditional on $F = -1.0104$ are $3.8344s$, $0.1171s$, and 0.1405, where s is the spread. Similar calculations are carried out for the other 59 factor values and equation (24.12) is used to integrate over F. The unconditional present values of payoffs, regular payments, and accrual payments are 0.0637, $4.0543s$, and $0.0531s$. The breakeven CDS spread is therefore $0.0637/(4.0543 + 0.0531) = 0.0155$, or 155 basis points.

Implied Correlation

In the standard market model, the recovery rate R is usually assumed to be 40%. This leaves the copula correlation ρ as the only unknown parameter. This makes the model similar to Black–Scholes–Merton, where there is only one unknown parameter, the volatility. Market participants like to imply a correlation from the market quotes for tranches in the same way that they imply a volatility from the market prices of options.

Suppose that the values of $\{\alpha_L, \alpha_H\}$ for successively more senior tranches are $\{\alpha_0, \alpha_1\}$, $\{\alpha_1, \alpha_2\}$, $\{\alpha_2, \alpha_3\}, \dots$, with $\alpha_0 = 0$. (For example, in the case of iTraxx Europe, $\alpha_0 = 0$, $\alpha_1 = 0.03$, $\alpha_2 = 0.06$, $\alpha_3 = 0.09$, $\alpha_4 = 0.12$, $\alpha_5 = 0.22$, $\alpha_6 = 1.00$.) There are two alternative implied correlations measures. One is *compound correlation*. For a tranche $\{\alpha_{q-1}, \alpha_q\}$, this is the value of the correlation, ρ, that leads to the spread calculated from the model being the same as the spread in the market. It is found using an iterative search. The other is *base correlation*. For a particular value of α_q ($q \geqslant 1$), this is the value of ρ that leads to the $\{0, \alpha_q\}$ tranche being priced consistently with the market. It is obtained using the following steps:

1. Calculate the compound correlation for each tranche.
2. Use the compound correlation to calculate the present value of the expected loss on each tranche during the life of the CDO as a percent of the initial tranche principal. This is the variable we have defined as C above. Suppose that the value of C for the α_{q-1} to α_q tranche is C_q.
3. Calculate the present value of the expected loss on the $\{0, \alpha_q\}$ tranche as a percent of the total principal of the underlying portfolio. This is $\sum_{p=1}^{q} C_p(\alpha_p - \alpha_{p-1})$.
4. The C-value for the $\{0, \alpha_q\}$ tranche is the value calculated in Step 3 divided by α_q. The base correlation is the value of the correlation parameter, ρ, that is consistent with this C-value. It is found using an iterative search.

The present value of the loss as a percent of underlying portfolio that would be calculated in Step 3 for the iTraxx Europe quotes for January 31, 2007, given in Table 24.6 are shown in Figure 24.3. The implied correlations for these quotes are shown in Table 24.8. The calculations were carried out using DerivaGem assuming that the term structure of interest rates is flat at 3% and the recovery rate is 40%. The CDSs worksheet shows that the 23-basis-point spread implies a hazard rate of 0.382%. The implied correlations are calculated using the CDOs worksheet. The values underlying Figure 24.3 can also be calculated with this worksheet using the expression in Step 3 above.

The correlation patterns in Table 24.8 are typical of those usually observed. The compound correlations exhibit a "correlation smile". As the tranche becomes more

Figure 24.3 Present value of expected loss on 0 to X% tranche as a percent of total underlying principal for iTraxx Europe on January 31, 2007.

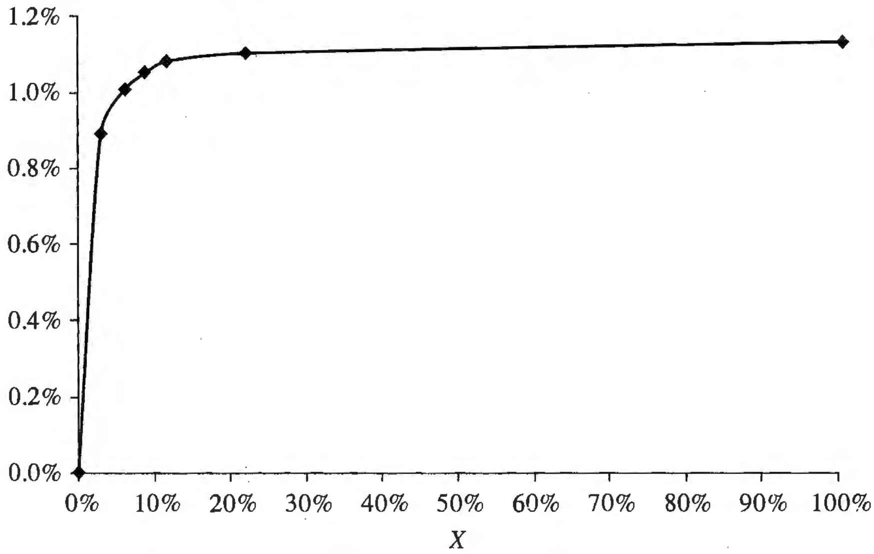


Table 24.8 Implied correlations for 5-year iTraxx Europe tranches on January 31, 2007.

Compound correlations					
Tranche	0–3%	3–6%	6–9%	9–12%	12–22%
Implied correlation	17.7%	7.8%	14.0%	18.2%	23.3%
Base correlations					
Tranche	0–3%	0–6%	0–9%	0–12%	0–22%
Implied correlation	17.7%	28.4%	36.5%	43.2%	60.5%

senior, the implied correlation first decreases and then increases. The base correlations exhibit a correlation skew where the implied correlation is an increasing function of the tranche detachment point.

If market prices were consistent with the one-factor Gaussian copula model, then the implied correlations (both compound and base) would be the same for all tranches. From the pronounced smiles and skews that are observed in practice, we can infer that market prices are not consistent with this model.

Valuing Nonstandard Tranches

We do not need a model to value the standard tranches of a standard portfolio such as iTraxx Europe because the spreads for these tranches can be observed in the market. Sometimes quotes need to be produced for nonstandard tranches of a standard portfolio. Suppose that you need a quote for the 4–8% iTraxx Europe tranche. One approach is to interpolate base correlations so as to estimate the base correlation for the 0–4% tranche and the 0–8% tranche. These two base correlations allow the present value of expected loss (as a percent of the underlying portfolio principal) to be estimated for these tranches. The present value of the expected loss for the 4–8% tranche (as a percent of the underlying principal) can be estimated as the difference between the present value of expected losses for the 0–8% and 0–4% tranches. This can be used to imply a compound correlation and a breakeven spread for the tranche.

It is now recognized that this is not the best way to proceed. A better approach is to calculate expected losses for each of the standard tranches and produce a chart such as Figure 24.3 showing the variation of expected loss for the 0–X% tranche with X. Values on this chart can be interpolated to give the expected loss for the 0–4% and the 0–8% tranches. The difference between these expected losses is a better estimate of the expected loss on the 4–8% tranche than that obtained from the base correlation approach.

It can be shown that for no arbitrage the expected losses in Figure 24.4 must increase at a decreasing rate. If base correlations are interpolated and then used to calculate expected losses, this no-arbitrage condition is often not satisfied. (The problem here is that the base correlation for the 0–X% tranche is a nonlinear function of the expected loss on the 0–X% tranche.) The direct approach of interpolating expected losses is therefore much better than the indirect approach of interpolating base correlations. What is more, it can be done so as to ensure that the no-arbitrage condition just mentioned is satisfied.

24.11 ALTERNATIVES TO THE STANDARD MARKET MODEL

This section outlines a number of alternatives to the one-factor Gaussian copula model that has become the market standard.

Heterogeneous Model

The standard market model is a homogeneous model in the sense that the time-to-default probability distributions are assumed to be the same for all companies and the copula correlations for any pair of companies are the same. The homogeneity assumption can be relaxed so that a more general model is used. However, this model is more complicated to implement because each company has a different probability of defaulting by any given time and $P(k, t \mid F)$ can no longer be calculated using the binomial formula in equation (24.7). It is necessary to use a numerical procedure such as that described in Andersen *et al.* (2003) and Hull and White (2004).[12]

Other Copulas

The one-factor Gaussian copula model is a particular model of the correlation between times to default. Many other one-factor copula models have been proposed. These include the Student t copula, the Clayton copula, Archimedean copula, and Marshall–Olkin copula. We can also create new one-factor copulas by assuming that F and the Z_i in equation (23.10) have nonnormal distributions with mean 0 and standard deviation 1. Hull and White show that a good fit to the market is obtained when F and the Z_i have Student t distributions with four degrees of freedom.[13] They call this the *double t copula.*

Another approach is to increase the number of factors in the model. Unfortunately, the model is then much slower to run because it is necessary to integrate over several normal distributions instead of just one.

Random Factor Loadings

Andersen and Sidenius have suggested a model where the copula correlation ρ in equation (24.5) is a function of F.[14]

In general, ρ increases as F decreases. This means that in states of the world where the default rate is high (i.e., states of the world where F is low) the default correlation is also high. There is empirical evidence suggesting that this is the case.[15] Andersen and Sidenius find that this model fits market quotes much better than the standard market model.

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[12] See L. Andersen, J. Sidenius, and S. Basu, "All Your Hedges in One Basket," *Risk*, November 2003; and J.C. Hull and A. White, "Valuation of a CDO and nth-to-Default Swap without Monte Carlo Simulation," *Journal of Derivatives*, 12, 2 (Winter 2004), 8–23.

[13] See J.C. Hull and A. White, "Valuation of a CDO and nth-to-Default Swap without Monte Carlo Simulation," *Journal of Derivatives*, 12, 2 (Winter 2004), 8–23.

[14] See L. Andersen and J. Sidenius, "Extension of the Gaussian Copula Model: Random Recovery and Random Factor Loadings," *Journal of Credit Risk*, 1, 1 (Winter 2004), 29–70.

[15] See, for example, A. Sevigny and O. Renault, "Default Correlation: Empirical Evidence," Working Paper, Standard and Poors, 2002; S.R. Das, L. Freed, G. Geng, and N. Kapadia, "Correlated Default Risk," *Journal of Fixed Income*, 16 (2006), 2, 7–32, J.C. Hull, M. Predescu, and A. White, "The Valuation of Correlation-Dependent Credit Derivatives Using a Structural Model," *Journal of Credit Risk*, 6 (2010), 99–132; and A. Ang and J. Chen, "Asymmetric Correlation of Equity Portfolios," *Journal of Financial Economics*, 63 (2002), 443–494.

The Implied Copula Model

Hull and White show how a copula can be implied from market quotes.[16] The simplest version of the model assumes that a certain average hazard rate applies to all companies in a portfolio over the life of a CDO. That average hazard rate has a probability distribution that can be implied from the pricing of tranches. The calculation of the implied copula is similar in concept to the idea, discussed in Chapter 19, of calculating an implied probability distribution for a stock price from option prices.

Dynamic Models

The models discussed so far can be characterized as static models. In essence they model the average default environment over the life of the CDO. The model constructed for a 5-year CDO is different from the model constructed for a 7-year CDO, which is in turn different from the model constructed for a 10-year CDO. Dynamic models are different from static models in that they attempt to model the evolution of the loss on a portfolio through time. There are three different types of dynamic models:

1. *Structural Models*: These are similar to the models described in Section 23.6 except that the stochastic processes for the asset prices of many companies are modeled simultaneously. When the asset price for a company reaches a barrier, there is a default. The processes followed by the assets are correlated. The problem with these types of models is that they have to be implemented with Monte Carlo simulation and calibration is therefore difficult.
2. *Reduced Form Models*: In these models the hazard rates of companies are modeled. In order to build in a realistic amount of correlation, it is necessary to assume that there are jumps in the hazard rates.
3. *Top Down Models*: These are models where the total loss on a portfolio is modeled directly. The models do not consider what happens to individual companies.

SUMMARY

Credit derivatives enable banks and other financial institutions to actively manage their credit risks. They can be used to transfer credit risk from one company to another and to diversify credit risk by swapping one type of exposure for another.

The most common credit derivative is a credit default swap. This is a contract where one company buys insurance from another company against a third company (the reference entity) defaulting on its obligations. The payoff is usually the difference between the face value of a bond issued by the reference entity and its value immediately after a default. Credit default swaps can be analyzed by calculating the present value of the expected payments and the present value of the expected payoff in a risk-neutral world.

[16] See J. C. Hull and A. White, "Valuing Credit Derivatives Using an Implied Copula Approach," *Journal of Derivatives*, 14 (2006), 8–28; and J. C. Hull and A. White, "An Improved Implied Copula Model and its Application to the Valuation of Bespoke CDO Tranches," *Journal of Investment Management*, 8, 3 (2010), 11–31.

A forward credit default swap is an obligation to enter into a particular credit default swap on a particular date. A credit default swap option is the right to enter into a particular credit default swap on a particular date. Both instruments cease to exist if the reference entity defaults before the date. A kth-to-default CDS is defined as a CDS that pays off when the kth default occurs in a portfolio of companies.

A total return swap is an instrument where the total return on a portfolio of credit-sensitive assets is exchanged for LIBOR plus a spread. Total return swaps are often used as financing vehicles. A company wanting to purchase a portfolio of assets will approach a financial institution to buy the assets on its behalf. The financial institution then enters into a total return swap with the company where it pays the return on the assets to the company and receives LIBOR plus a spread. The advantage of this type of arrangement is that the financial institution reduces its exposure to defaults by the company.

In a collateralized debt obligation a number of different securities are created from a portfolio of corporate bonds or commercial loans. There are rules for determining how credit losses are allocated. The result of the rules is that securities with both very high and very low credit ratings are created from the portfolio. A synthetic collateralized debt obligation creates a similar set of securities from credit default swaps. The standard market model for pricing both a kth-to-default CDS and tranches of a synthetic CDO is the one-factor Gaussian copula model for time to default.

FURTHER READING

Andersen, L., and J. Sidenius, "Extensions to the Gaussian Copula: Random Recovery and Random Factor Loadings," *Journal of Credit Risk*, 1, No. 1 (Winter 2004): 29–70.

Andersen, L., J. Sidenius, and S. Basu, "All Your Hedges in One Basket," *Risk*, November 2003.

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Schönbucher, P. J., *Credit Derivatives Pricing Models*. New York: Wiley, 2003.

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Practice Questions (Answers in Solutions Manual)

24.1. Explain the difference between a regular credit default swap and a binary credit default swap.

24.2. A credit default swap requires a semiannual payment at the rate of 60 basis points per year. The principal is $300 million and the credit default swap is settled in cash. A default occurs after 4 years and 2 months, and the calculation agent estimates that the price of the cheapest deliverable bond is 40% of its face value shortly after the default. List the cash flows and their timing for the seller of the credit default swap.

24.3. Explain the two ways a credit default swap can be settled.

24.4. Explain how a cash CDO and a synthetic CDO are created.

24.5. Explain what a first-to-default credit default swap is. Does its value increase or decrease as the default correlation between the companies in the basket increases? Explain.

24.6. Explain the difference between risk-neutral and real-world default probabilities.

24.7. Explain why a total return swap can be useful as a financing tool.

24.8. Suppose that the risk-free zero curve is flat at 7% per annum with continuous compounding and that defaults can occur halfway through each year in a new 5-year credit default swap. Suppose that the recovery rate is 30% and the default probabilities each year conditional on no earlier default is 3%. Estimate the credit default swap spread. Assume payments are made annually.

24.9. What is the value of the swap in Problem 24.8 per dollar of notional principal to the protection buyer if the credit default swap spread is 150 basis points?

24.10. What is the credit default swap spread in Problem 24.8 if it is a binary CDS?

24.11. How does a 5-year nth-to-default credit default swap work? Consider a basket of 100 reference entities where each reference entity has a probability of defaulting in each year of 1%. As the default correlation between the reference entities increases what would you expect to happen to the value of the swap when (a) $n = 1$ and (b) $n = 25$. Explain your answer.

24.12. What is the formula relating the payoff on a CDS to the notional principal and the recovery rate?

24.13. Show that the spread for a new plain vanilla CDS should be $(1 - R)$ times the spread for a similar new binary CDS, where R is the recovery rate.

24.14. Verify that, if the CDS spread for the example in Tables 24.1 to 24.4 is 100 basis points, the probability of default in a year (conditional on no earlier default) must be 1.61%. How does the probability of default change when the recovery rate is 20% instead of 40%? Verify that your answer is consistent with the implied probability of default being approximately proportional to $1/(1 - R)$, where R is the recovery rate.

24.15. A company enters into a total return swap where it receives the return on a corporate bond paying a coupon of 5% and pays LIBOR. Explain the difference between this and a regular swap where 5% is exchanged for LIBOR.

24.16. Explain how forward contracts and options on credit default swaps are structured.

24.17. "The position of a buyer of a credit default swap is similar to the position of someone who is long a risk-free bond and short a corporate bond." Explain this statement.

24.18. Why is there a potential asymmetric information problem in credit default swaps?

24.19. Does valuing a CDS using real-world default probabilities rather than risk-neutral default probabilities overstate or understate its value? Explain your answer.

24.20. What is the difference between a total return swap and an asset swap?

24.21. Suppose that in a one-factor Gaussian copula model the 5-year probability of default for each of 125 names is 3% and the pairwise copula correlation is 0.2. Calculate, for factor values of −2, −1, 0, 1, and 2: (a) the default probability conditional on the factor value and (b) the probability of more than 10 defaults conditional on the factor value.

24.22. Explain the difference between base correlation and compound correlation.

24.23. In Example 24.2, what is the tranche spread for the 9% to 12% tranche?

Further Questions

24.24. Suppose that the risk-free zero curve is flat at 6% per annum with continuous compounding and that defaults can occur at times 0.25 years, 0.75 years, 1.25 years, and 1.75 years in a 2-year plain vanilla credit default swap with semiannual payments. Suppose that the recovery rate is 20% and the unconditional probabilities of default (as seen at time zero) are 1% at times 0.25 years and 0.75 years, and 1.5% at times 1.25 years and 1.75 years. What is the credit default swap spread? What would the credit default spread be if the instrument were a binary credit default swap?

24.25. Assume that the default probability for a company in a year, conditional on no earlier defaults is λ and the recovery rate is R. The risk-free interest rate is 5% per annum. Default always occurs halfway through a year. The spread for a 5-year plain vanilla CDS where payments are made annually is 120 basis points and the spread for a 5-year binary CDS where payments are made annually is 160 basis points. Estimate R and λ.

24.26. Explain how you would expect the returns offered on the various tranches in a synthetic CDO to change when the correlation between the bonds in the portfolio increases.

24.27. Suppose that:
(a) The yield on a 5-year risk-free bond is 7%.
(b) The yield on a 5-year corporate bond issued by company X is 9.5%.
(c) A 5-year credit default swap providing insurance against company X defaulting costs 150 basis points per year.

What arbitrage opportunity is there in this situation? What arbitrage opportunity would there be if the credit default spread were 300 basis points instead of 150 basis points? Give two reasons why arbitrage opportunities such as those you identify are less than perfect.

24.28. In Example 24.3, what is the spread for (a) a first-to-default CDS and (b) a second-to default CDS?

24.29. In Example 24.2, what is the tranche spread for the 6% to 9% tranche?

24.30. The 1-, 2-, 3-, 4-, and 5-year CDS spreads are 100, 120, 135, 145, and 152 basis points, respectively. The risk-free rate is 3% for all maturities, the recovery rate is 35%, and payments are quarterly. Use DerivaGem to calculate the continuously compounded hazard rate each year. What is the probability of default in year 1? What is the probability of default in year 2?

24.31. Table 24.6 shows the five-year iTraxx index was 77 basis points on January 31, 2008. Assume the risk-free rate is 5% for all maturities, the recovery rate is 40%, and payments are quarterly. Assume also that the spread of 77 basis points applies to all maturities. Use the DerivaGem CDS worksheet to calculate a hazard rate consistent with the spread. Use this in the CDO worksheet with 10 integration points to imply base correlations for each tranche from the quotes for January 31, 2008.